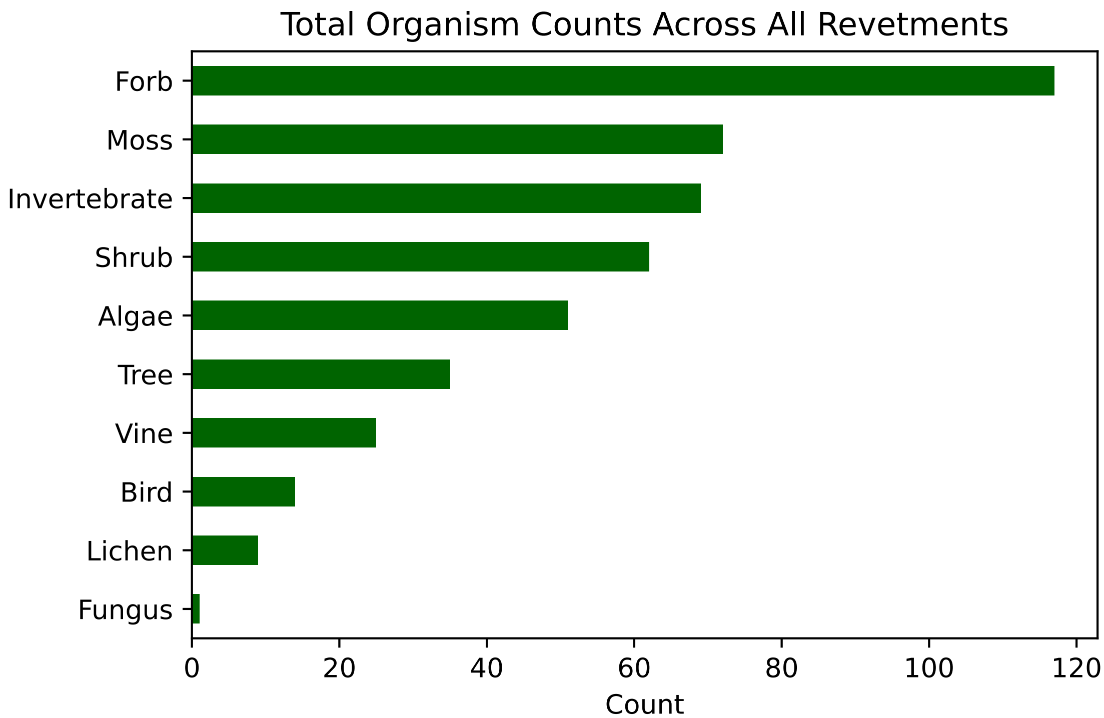
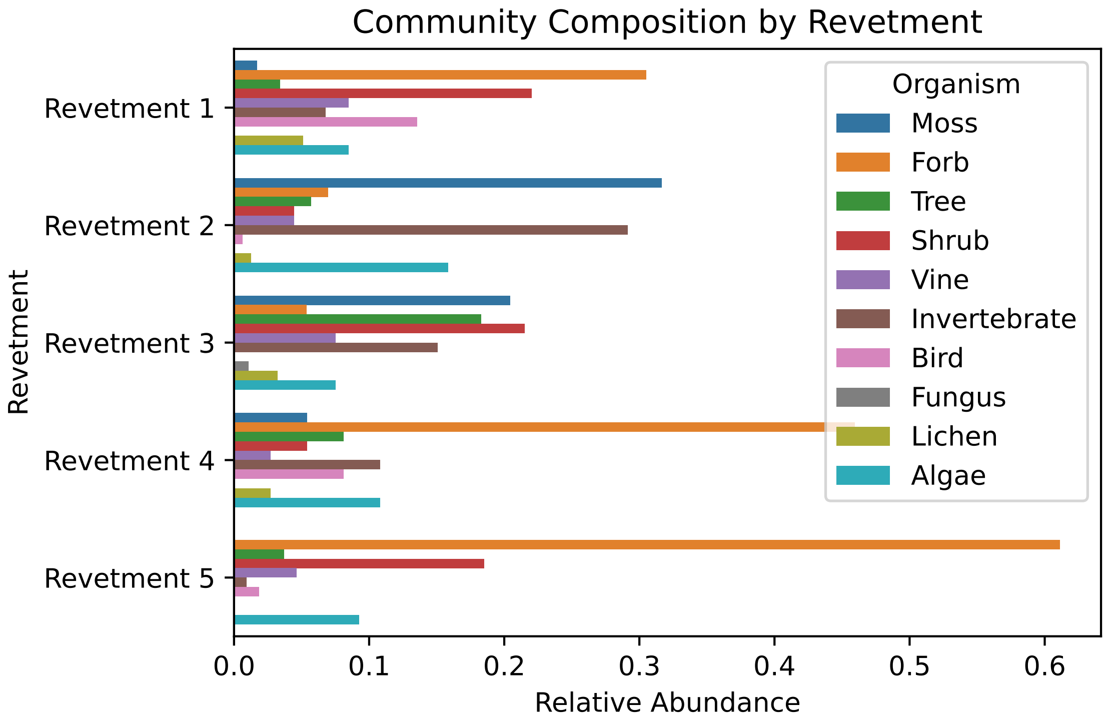
Revetment Dataset Biodiversity Visualizations

Bar Chart of Overall Organism Counts

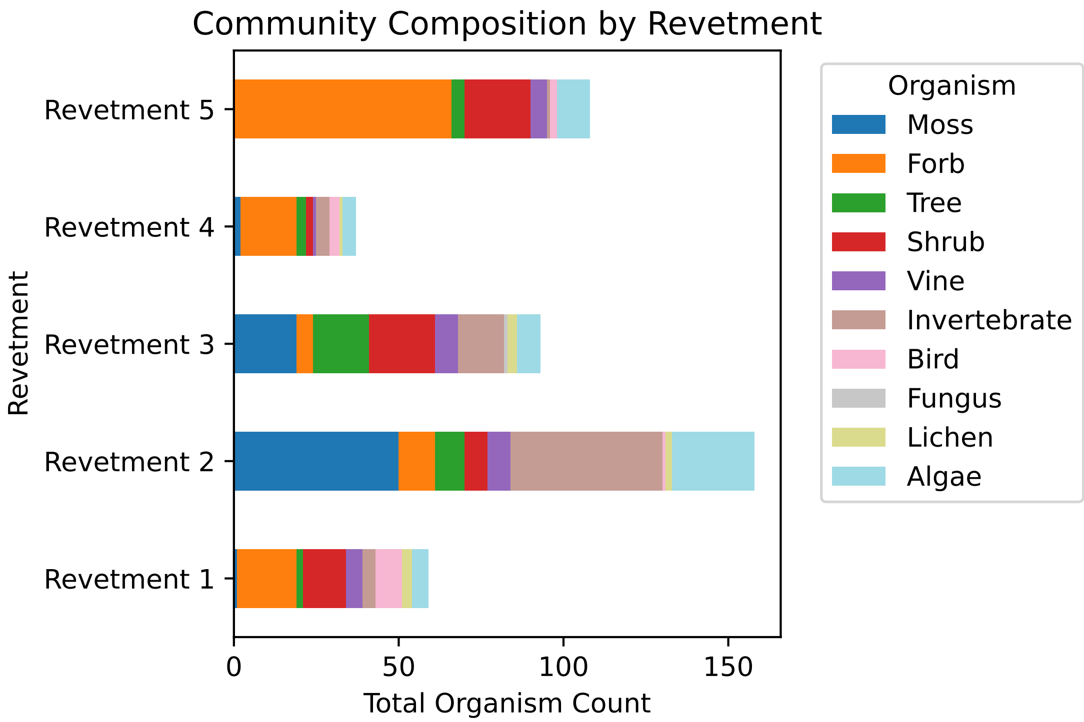


Total observed counts of each organism type across all revetments. Algae and Lichen are the most abundant.

Stacked Bar Chart of Organisms per Revetment

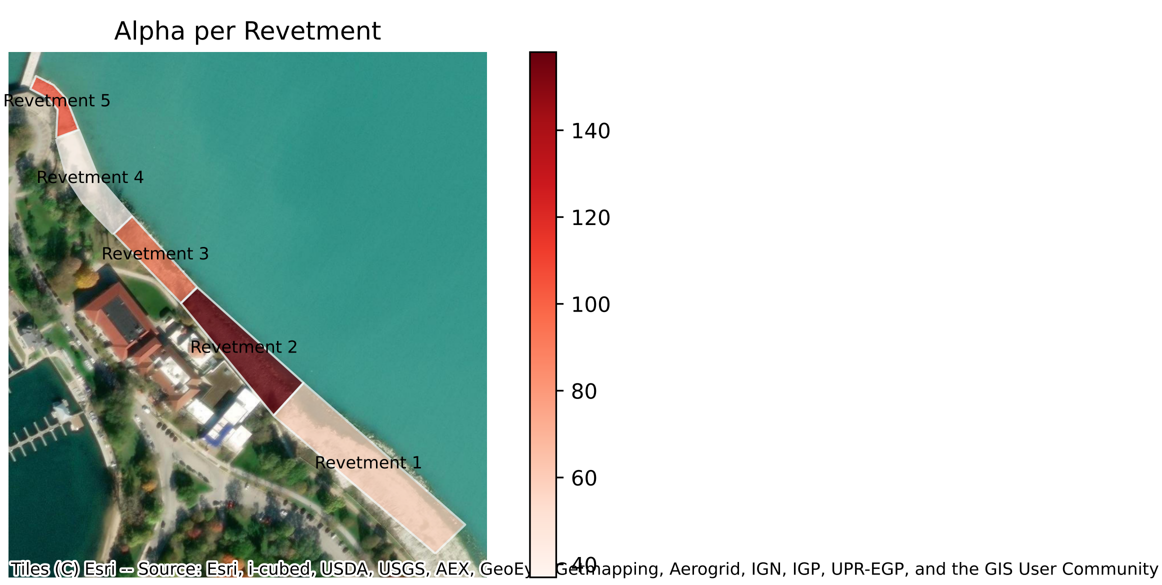


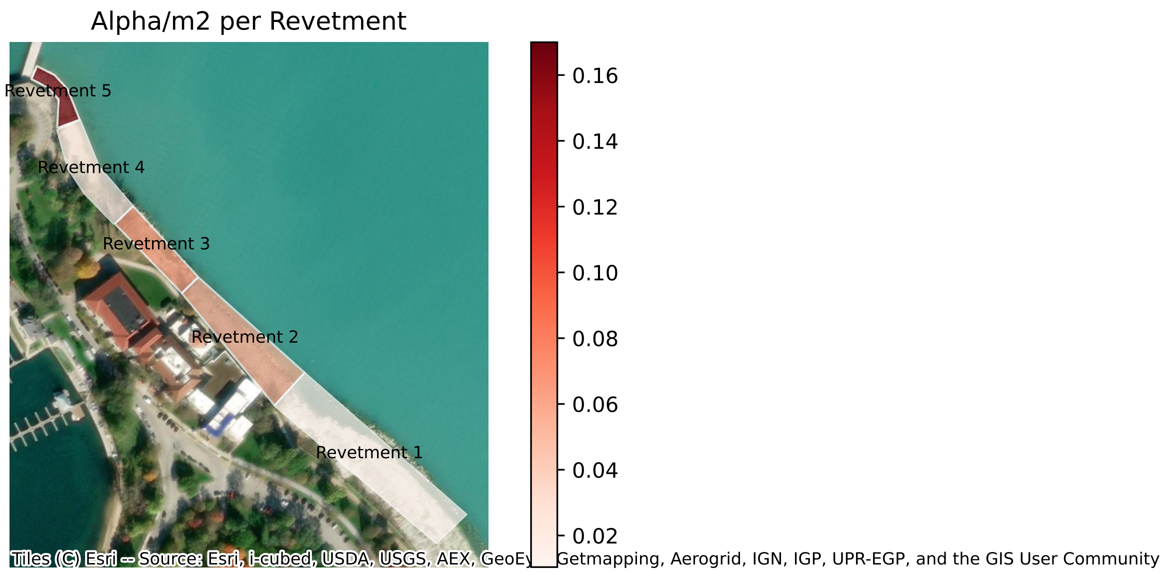
Relative organism composition within each revetment. Revetments 1 and 4 are more balanced across types, while others are dominated by a few organisms, possibly suggesting different ecological niches.

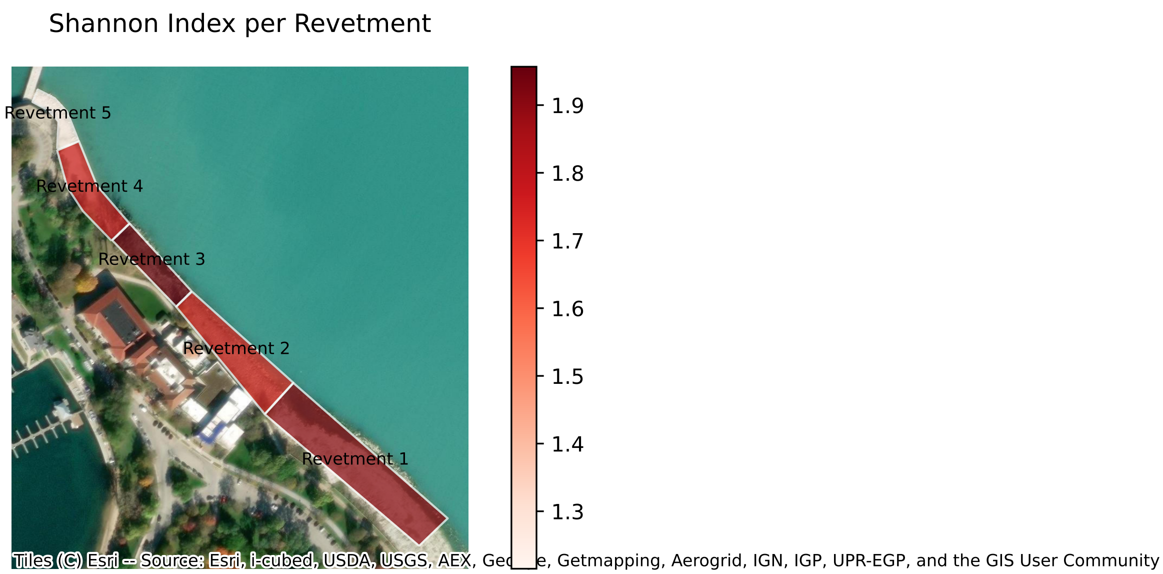


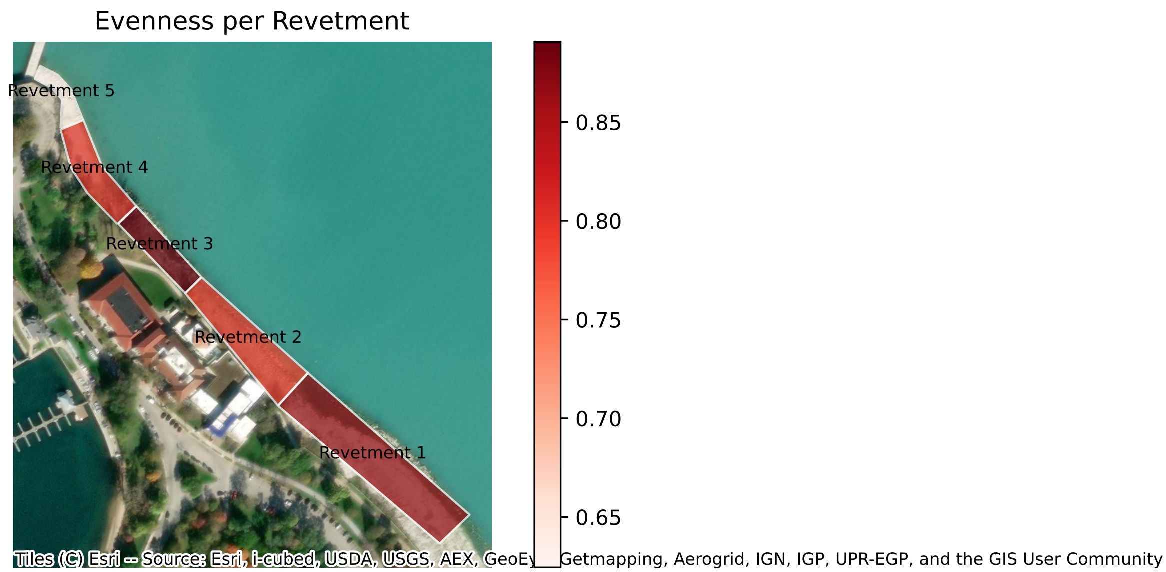
Need to figure out which version of this looks best.

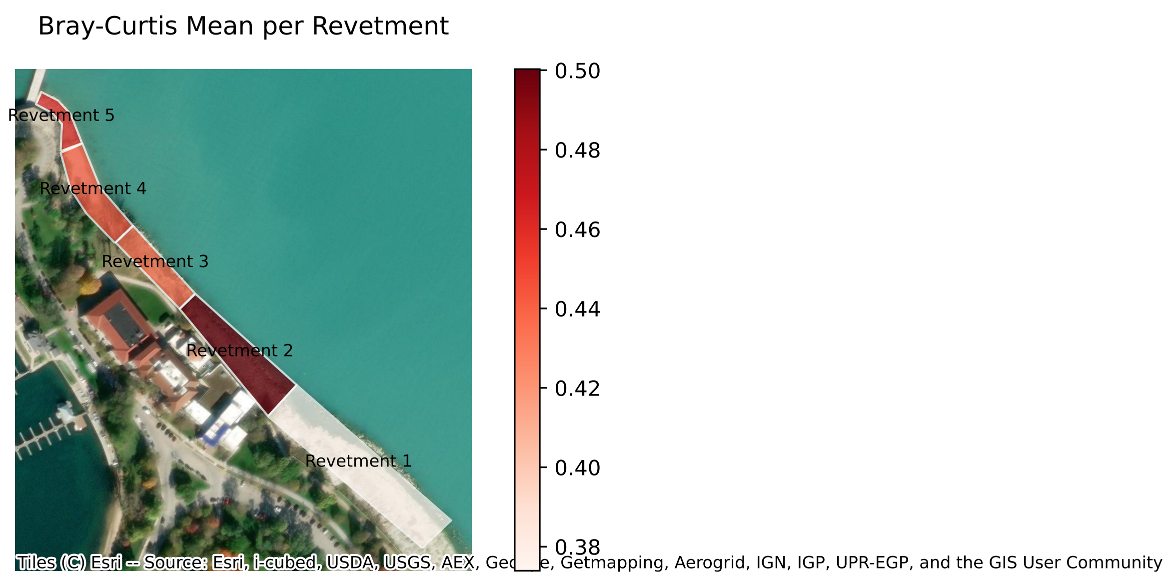
Choropleth Maps (Alpha, Alpha/m2, Shannon, Bray-Curtis, Evenness)




Alpha/m2 per Revetment
Revetment 5
Revetment 4
Revetment 3
Revetment 2
Revetment 1
0.16
0.14
0.12
0.10
0.08
0.06
0.04
0.02
Tiles (C) Esri -- Source: Esri, i-cubed, USDA, USGS, AEX, GeoEye, Getmapping, Aerogrid, IGN, IGP, UPR-EGP, and the GIS User Community


Shannon Index per Revetment
Revetment 5
Revetment 4
Revetment 3
Revetment 2
Revetment 1
1.9
1.8
1.7
1.6
1.5
1.4
1.3
Tiles (C) Esri -- Source: Esri, i-cubed, USDA, USGS, AEX, GeoEye, Getmapping, Aerogrid, IGN, IGP, UPR-EGP, and the GIS User Community





Alpha: Revetment 2 has the most distinct taxa.

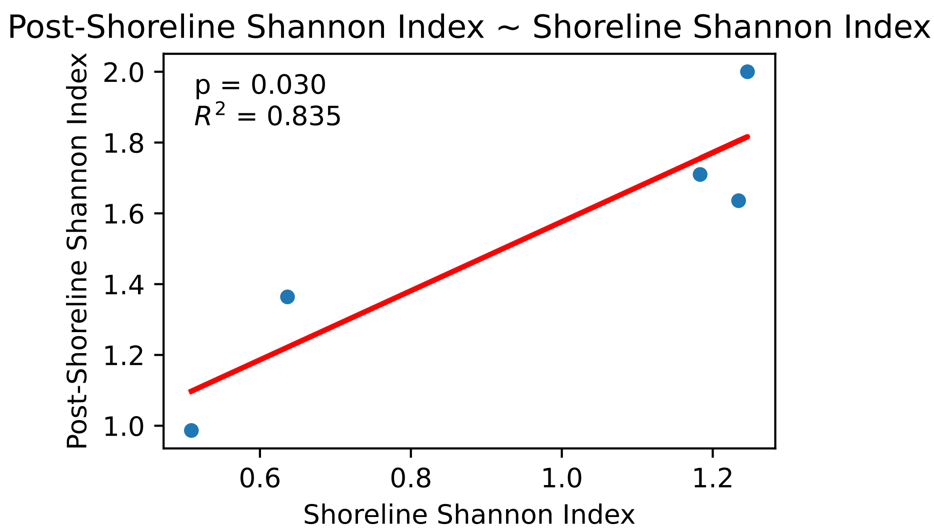
Alpha/m²: Biological density; shows revetments 5 as the highest.

Shannon: Community diversity; revetments 1 and 3 here.

Evenness: Balance in community; revetment 5 is highly uneven. Forb domination.

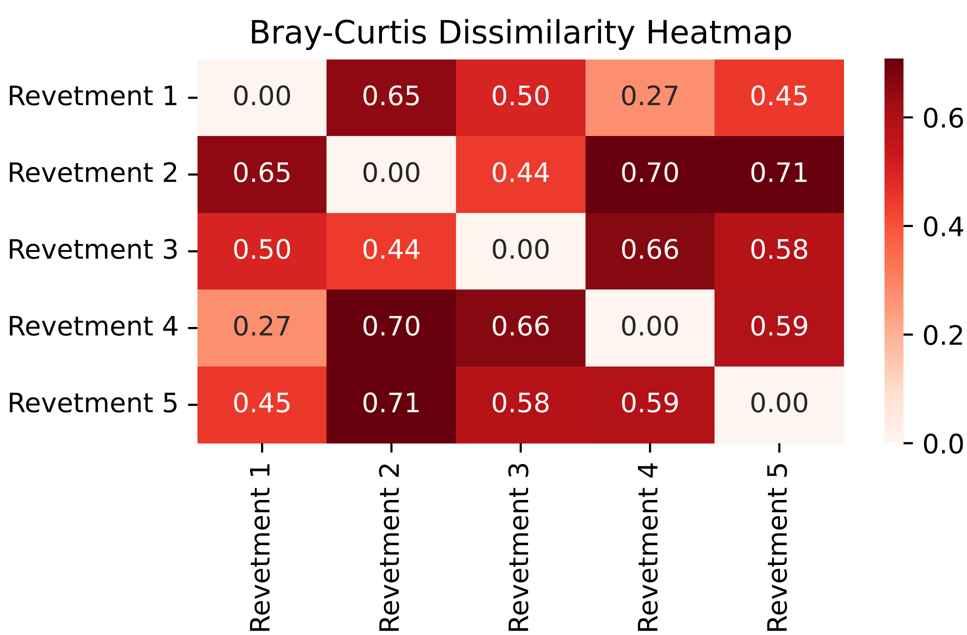
Bray-Curtis Mean: Beta diversity; revetment 2 is the most ecologically different.

Linear Regression: Shoreline and Post-Shoreline Shannon



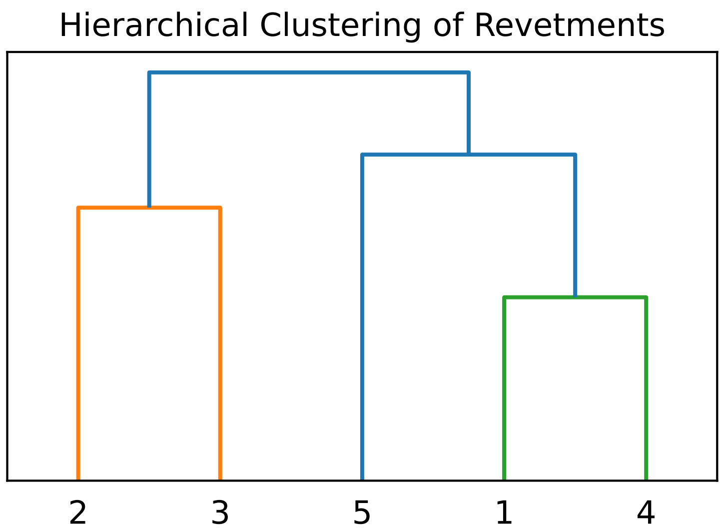
Shows that shoreline diversity may effect what comes later, could be interesting for successional reasons. Could help us talk about a predicted abundance like is mentioned on guidelines.

Bray-Curtis Heatmap



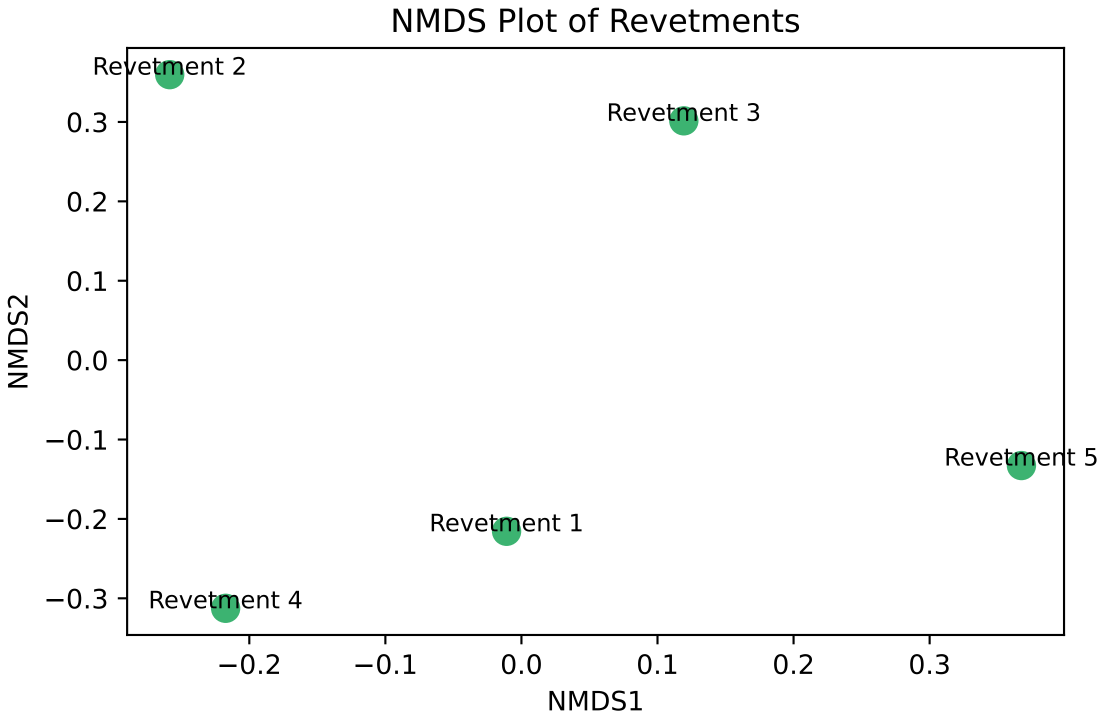
Kinda cool plot... shows how dissimilar revetments are to eachother, 1 = completely dissimilar, 0 = identical.

Hierarchical Clustering

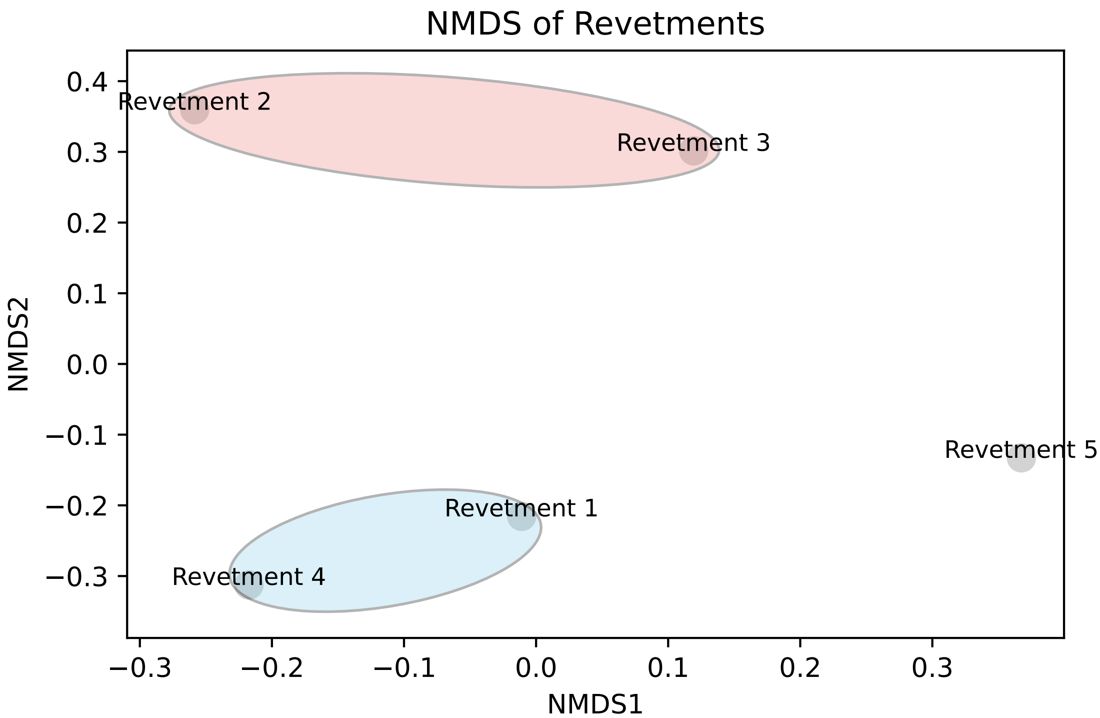


Clustering similar revetments based on the matrix. Could be cool to pair with the genetic phylogeny.

NMDS Plot

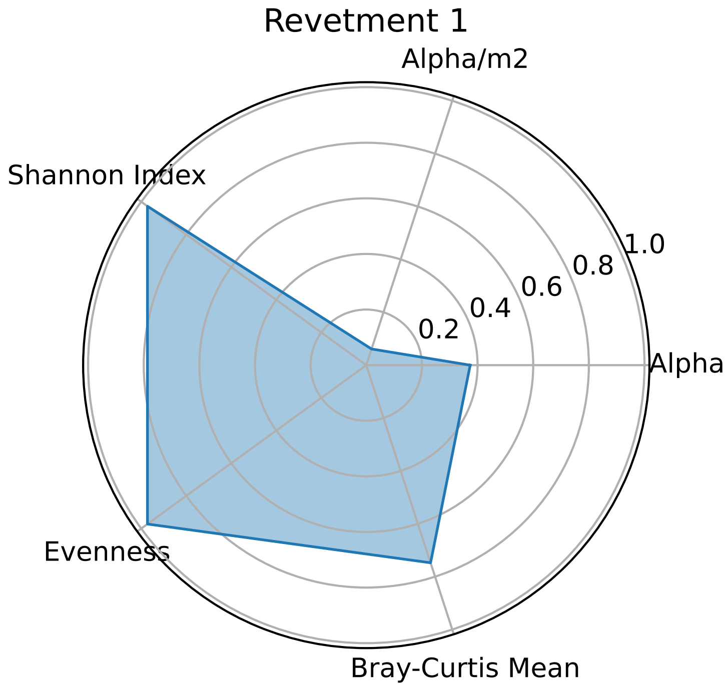


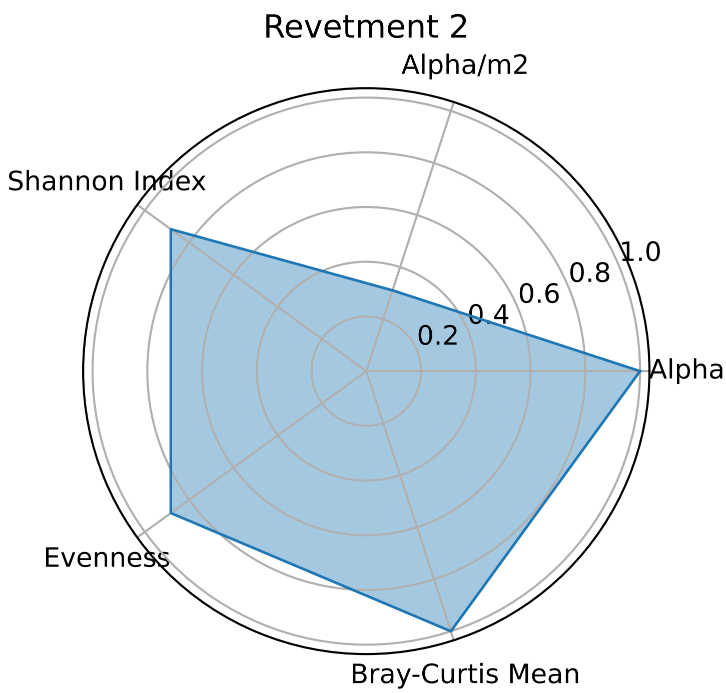
Got this idea from the bird reading from class... another way to show similarities between revetments. I see three regimes, 1 and 4, 2 and 3, and 5 by itself.

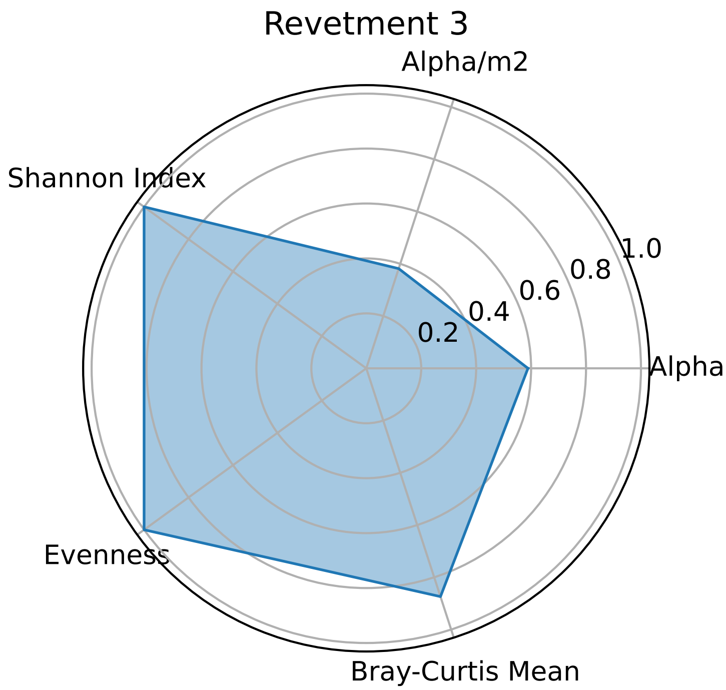


This one highlights the groupings/regimes.

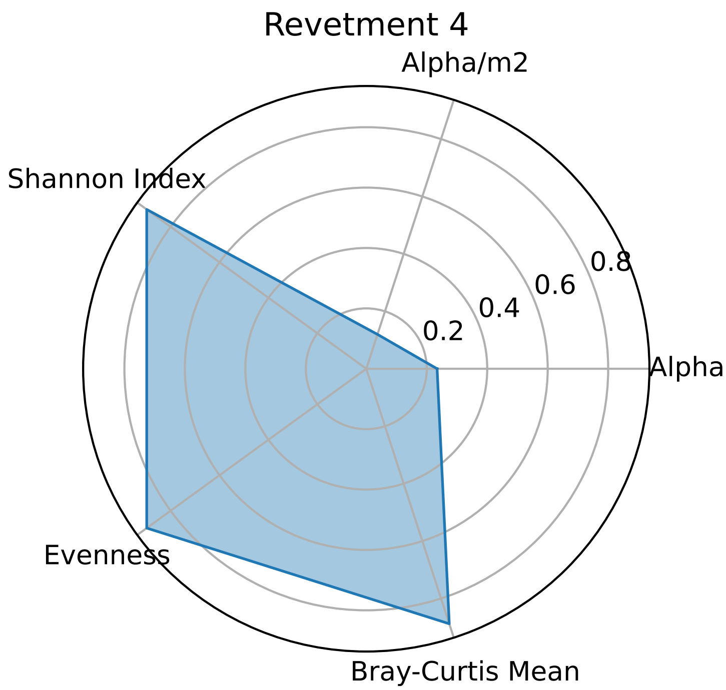
Radar Plots for Each Revetment



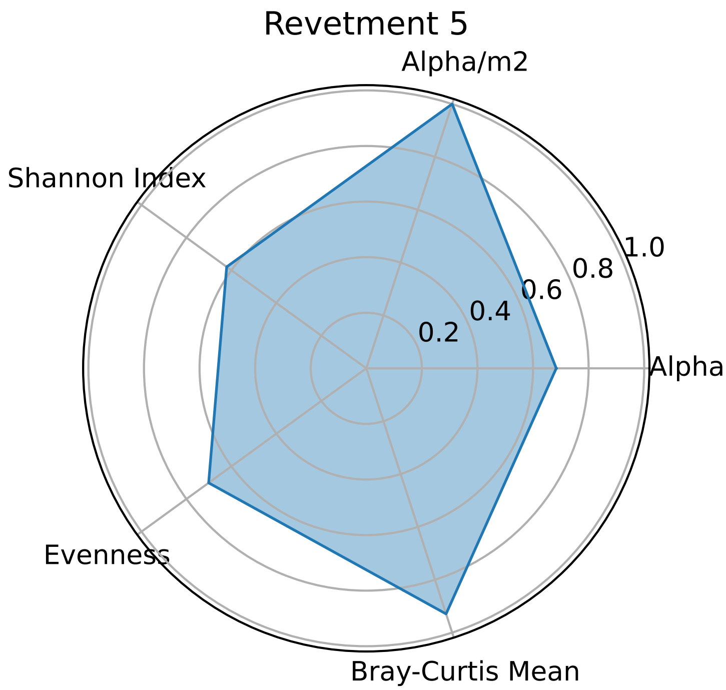




Revetment 3
Alpha/m2
Shannon Index
1.0
0.8
0.6
0.4
0.2
Alpha
Evenness
Bray-Curtis Mean



Revetment 4
Alpha/m2
Shannon Index
0.8
0.6
0.4
0.2
Alpha
Evenness
Bray-Curtis Mean



Cool way of visually comparing the metrics of each revetment, got this idea from a game. Helps visually how similar revetments are and supports the regimes from above, really shows how different revetment 5 is.

Stuff to Add

- Metrics on revetment:
  - Effectiveness
  - Cost
  - Material
  - An overall scoring of which revetment (if different enough, I am not sure there is any significant difference in biodiversity between them, there is a difference between them abundance wise for sure and how they may promote that abundance.) is best when looking at all metrics, with an emphasis on preserving biodiversity(? Could be a worthwile conclusion/suggestion that I think we are capable of making with our data)